

Wipro NGA .NET Library Management System - NUnit Code

Library Management System Classes + NUnit Tests

Book.cs

```
-----  
public class Book  
{  
    public string Title { get; set; }  
    public string Author { get; set; }  
    public string ISBN { get; set; }  
    public bool IsBorrowed { get; private set; }  
  
    public Book(string title, string author, string isbn)  
    {  
        Title = title;  
        Author = author;  
        ISBN = isbn;  
        IsBorrowed = false;  
    }  
  
    public void Borrow()  
    {  
        if (IsBorrowed) throw new Exception("Already borrowed");  
        IsBorrowed = true;  
    }  
  
    public void Return()  
    {  
        IsBorrowed = false;  
    }  
}
```

Borrower.cs

```
-----  
using System.Collections.Generic;  
  
public class Borrower  
{  
    public string Name { get; set; }  
    public string LibraryCardNumber { get; set; }  
    public List<Book> BorrowedBooks { get; set; } = new List<Book>();  
  
    public Borrower(string name, string card)  
    {  
        Name = name;  
        LibraryCardNumber = card;  
    }  
  
    public void BorrowBook(Book book)  
    {  
        BorrowedBooks.Add(book);  
    }  
  
    public void ReturnBook(Book book)  
    {  
        BorrowedBooks.Remove(book);  
    }  
}
```

Library.cs

```
-----  
using System.Collections.Generic;  
using System.Linq;  
  
public class Library  
{  
    public List<Book> Books = new List<Book>();  
    public List<Borrower> Borrowers = new List<Borrower>();  
  
    public void AddBook(Book book) => Books.Add(book);  
    public void RegisterBorrower(Borrower b) => Borrowers.Add(b);  
}
```

```

    public void BorrowBook(string isbn, string card)
    {
        var book = Books.First(b => b.ISBN == isbn);
        var borrower = Borrowers.First(b => b.LibraryCardNumber == card);

        book.Borrow();
        borrower.BorrowBook(book);
    }

    public void ReturnBook(string isbn, string card)
    {
        var book = Books.First(b => b.ISBN == isbn);
        var borrower = Borrowers.First(b => b.LibraryCardNumber == card);

        book.Return();
        borrower.ReturnBook(book);
    }

    public List<Book> ViewBooks() => Books;
    public List<Borrower> ViewBorrowers() => Borrowers;
}

```

NUnit Tests

using NUnit.Framework;

```

[TestFixture]
public class LibraryTests
{
    private Library library;
    private Book book;
    private Borrower borrower;

    [SetUp]
    public void Setup()
    {
        library = new Library();
        book = new Book("Clean Code", "Robert", "111");
        borrower = new Borrower("Tushar", "CARD1");

        library.AddBook(book);
        library.RegisterBorrower(borrower);
    }

    [Test]
    public void AddBook_Test()
    {
        library.AddBook(new Book("Java", "James", "222"));
        Assert.That(library.Books.Count, Is.EqualTo(2));
    }

    [Test]
    public void Borrow_Test()
    {
        library.BorrowBook("111", "CARD1");
        Assert.That(book.IsBorrowed, Is.True);
    }

    [Test]
    public void Return_Test()
    {
        library.BorrowBook("111", "CARD1");
        library.ReturnBook("111", "CARD1");
        Assert.That(book.IsBorrowed, Is.False);
    }
}

```